Binomial Theorem

Short Answer Questions

1. Expand $(x+2)^5$ using the Binomial Theorem.
2. Find the coefficient of x^4 in the expansion of $(2x-3)^6$.
3. Find the middle term in the expansion of $\left(x+\frac{1}{x}\right)^8$.
4. Find the term independent of x in the expansion of $\left(x^2+\frac{1}{x}\right)^9$.
5. In the expansion of $(1-2x)^7$, find the coefficient of x^3.
6. Find the 5th term in the expansion of $(2x-1)^6$.
7. Find the general term in the expansion of $\left(x-\frac{3}{x}\right)^7$.
8. Find the coefficient of x^5 in the expansion of $\left(x^2-\frac{1}{x}\right)^6$.
9. Find the middle term of the expansion of $(1+x)^9$.
10. In the expansion of $(a+bx)^n$, find the term containing x^r.

Long Answer Questions

11. Prove that the sum of coefficients in $(1+x)^n$ is 2^n.
12. If the 5th and 6th terms in the expansion of $(1+x)^n$ are equal, find the value of n.
13. Show that the middle term of $(1+x)^{2n}$ is ${}_{2n}C_n x^n$.
14. Find the term independent of x in the expansion of $\left(x^2+\frac{1}{x}\right)^9$.
15. Find the term containing x^3 in the expansion of $(2x^2-3x)^6$.
16. Find the value of ${}_{10}C_4 + {}_{10}C_5 + {}_{10}C_6$.
17. Prove that $\sum_{r=0}^{n} {}_nC_r = 2^n$.
18. Expand $(1-x)^5$ and verify for $x=\frac{1}{2}$.
19. Using the Binomial Theorem, prove that

$$(1+x)^4+(1-x)^4=2(1+6x^2+x^4)$$

20. Find the value of x if the ratio of the 4th term to the 5th term in the expansion of $(1+x)^n$ is $2:3$ and $n=7$.